AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 11 · WORKBOOK

Repeating Efficiently

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected loop record.

Trace the loop, test an initially false condition, reject an extra update and repair an unreachable target.

Keep one habit: Keep the body executions and final report separate.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

If/Then Decisions

Features: Useful Clues

Pictures Become Numbers

W1 · Trace every condition check

> Include the final false test and count body executions.

Your evidence record

- Start counter at 2. While counter is below 8, add 2. When false, report and stop. Record the value and result at every test, including the final false test. State the number of **iteration**s and the reported value.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Test zero iterations

> Run the first test before any body update.

Your evidence record

- Use the same below 8 **loop** with add 2. Make separate runs from 8 and 9. Explain how many times the **body** runs and what each run reports. Why would adding 2 once be an error?

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Find the first unauthorised update

> Keep the overshoot value and stop at its false test.

Your evidence record

- Start counter at 3. While counter is below 8, add 2. A learner writes values 3, 5, 7, 9, 11 and reports 11. Identify the first unauthorised update. Give the correct final test, reported value and **iteration** count.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Repair an unreachable equality target

> Use parity and a repaired boundary to justify stopping.

Your evidence record

- A procedure says start at 4; while counter is not equal to 9, add 2; report and stop. Write the first four values tested. Explain why the **loop** will not reach 9 using whole numbers.

  Suggest a changed condition that would stop once the counter reaches or passes 9, and trace that repaired run.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Transfer to a countdown

> Keep one first attempt before hints or feedback.

Read and explain independently

A paper countdown starts at 7. While remaining is above 1, subtract 2. When false, report and stop. Trace the run and count **iteration**s. Repeat from 6 and from 1. Explain why the final value need not equal 1.

What happens from 7 if subtract 2 is replaced by subtract 0?

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Transfer to a countdown

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Use a **loop** to **process** each item card in the test pile, with an explicit stop when no cards remain.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Check the first test for **Loop**s and stopping rules. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Loop**s and stopping rules. Proposed explanation: “Every **loop** runs at least once”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A robot stamps the next unstamped card until it has stamped five cards. Only three cards are available. Write a stopping rule that handles both limits and trace the number stamped.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Repeat take the next unstamped card and stamp it four times.

How many cards are stamped? What assumption matters?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Repeat clap then tap three times. Write the complete sequence. How many actions occur? What would happen if the stopping instruction were missing?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 11 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A robot stamps the next unstamped card until it has stamped five cards. Only three cards are available. Write a stopping rule that handles both limits and trace the number stamped.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A paper countdown starts at 7. While remaining is above 1, subtract 2. When false, report and stop. Trace the run and count **iteration**s. Repeat from 6 and from 1. Explain why the final value need not equal 1.

What happens from 7 if subtract 2 is replaced by subtract 0?

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional paper counter advances through a display until its condition becomes false.

Start at 2. While the current counter is below 10, add 3.

Test before each addition; report when the condition is false.

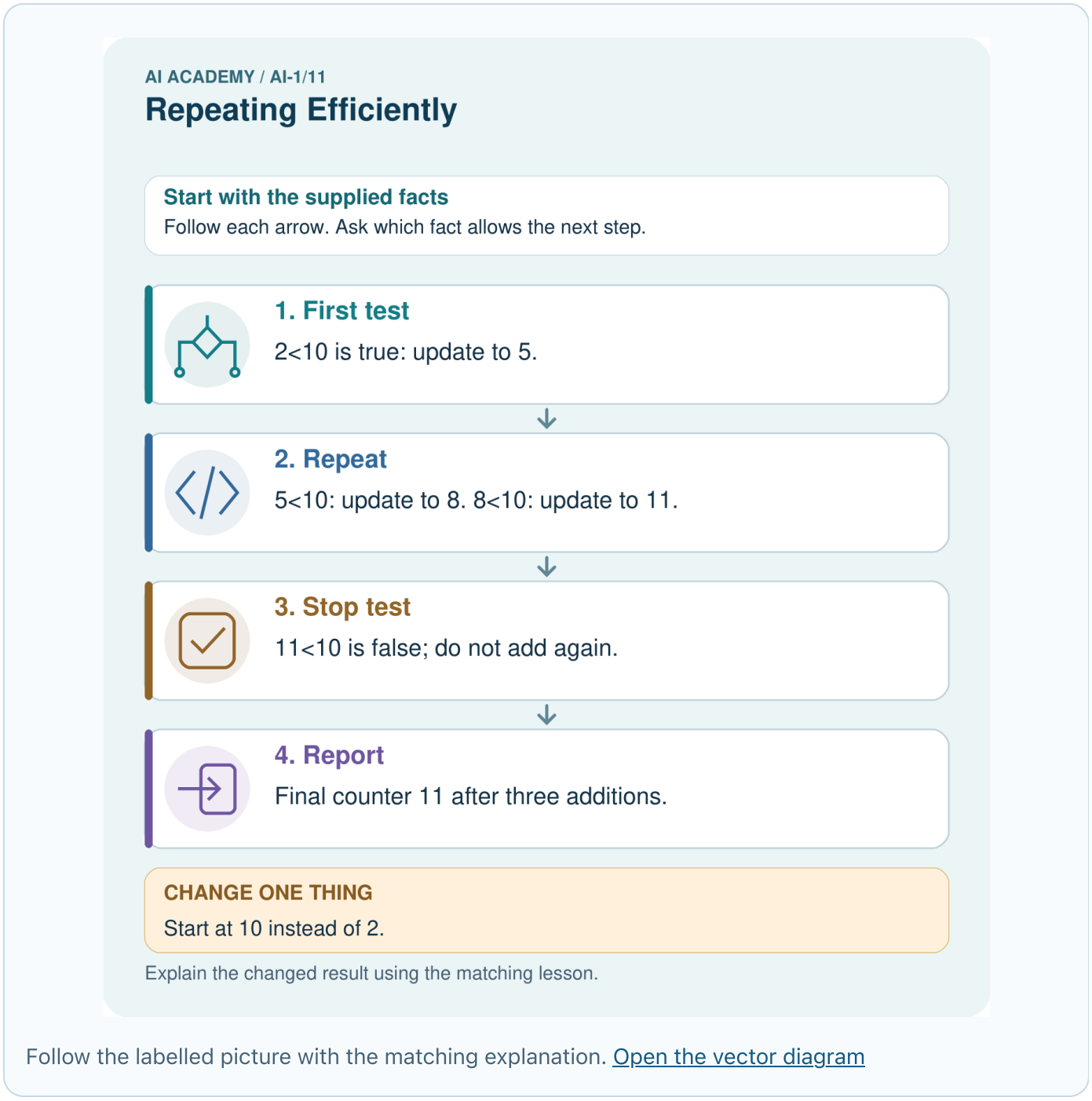
AI ACADEMY / AI-1/11

Condition checks before additions

	Current	Below 10?	Next
Check 1	2	true	5
Check 2	5	true	8
Check 3	8	true	11
Check 4	11	false	stop

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace a fresh start of 9. How many additions occur, and must the report equal 10?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Give any repeated sorter step an explicit **stopping condition** and trace a zero-**iteration** example.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Give any repeated sorter step an explicit **stopping condition** and trace a zero-**iteration** example.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional paper counter advances through a display until its condition becomes false.
- Which labelled step matches this detail? Final counter 11 after three additions.
- 1. Report
- 2. First test
- 3. Repeat
- 4. Stop test
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report Final counter 11 after three additions. The condition checks whether to do the next update; it does not constrain every resulting value to stay below 10.

The final 11 is therefore allowed by this procedure. Record both the condition and counter to distinguish a correct stop from an assumed endpoint.

Changed-case question

- Changed-case practice: Start at 10 instead of 2. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The **loop** runs zero times and reports 10.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The **loop** runs zero times and reports 10. Its first condition is already false, so no addition happens.

Repair and save your work

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which **input** detects the difference between the two written conditions?

Explain its entire run.. Return to this week and explain what you changed.

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a stated task.

Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.. Return to this week and explain what you changed.

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention.

Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Repeating Efficiently: the case

AI ACADEMY / AI-1/11 · 01 · LOOK AT THE FACTS

Repeating Efficiently: the case

The situation

A fictional paper counter advances through a display until its condition becomes false.

Supplied evidence

Start at 2. While the current counter is below 10, add 3.
Test before each addition; report when the condition is false.

By the end, I can check a loop condition before each update and count iterations.

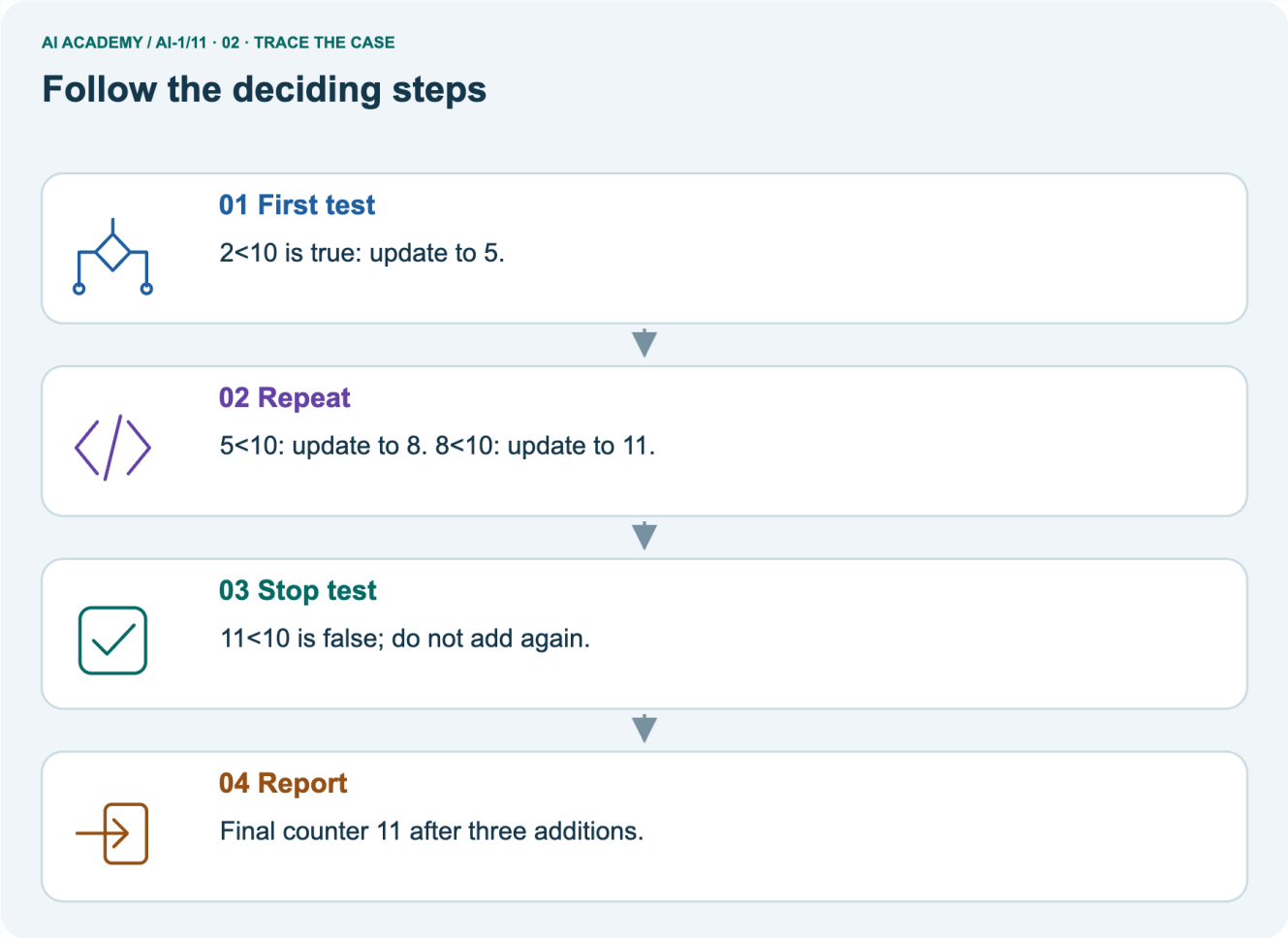
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper counter advances through a display until its condition becomes false. Supplied facts: Start at 2. While the current counter is below 10, add 3. Test before each addition; report when the condition is false.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



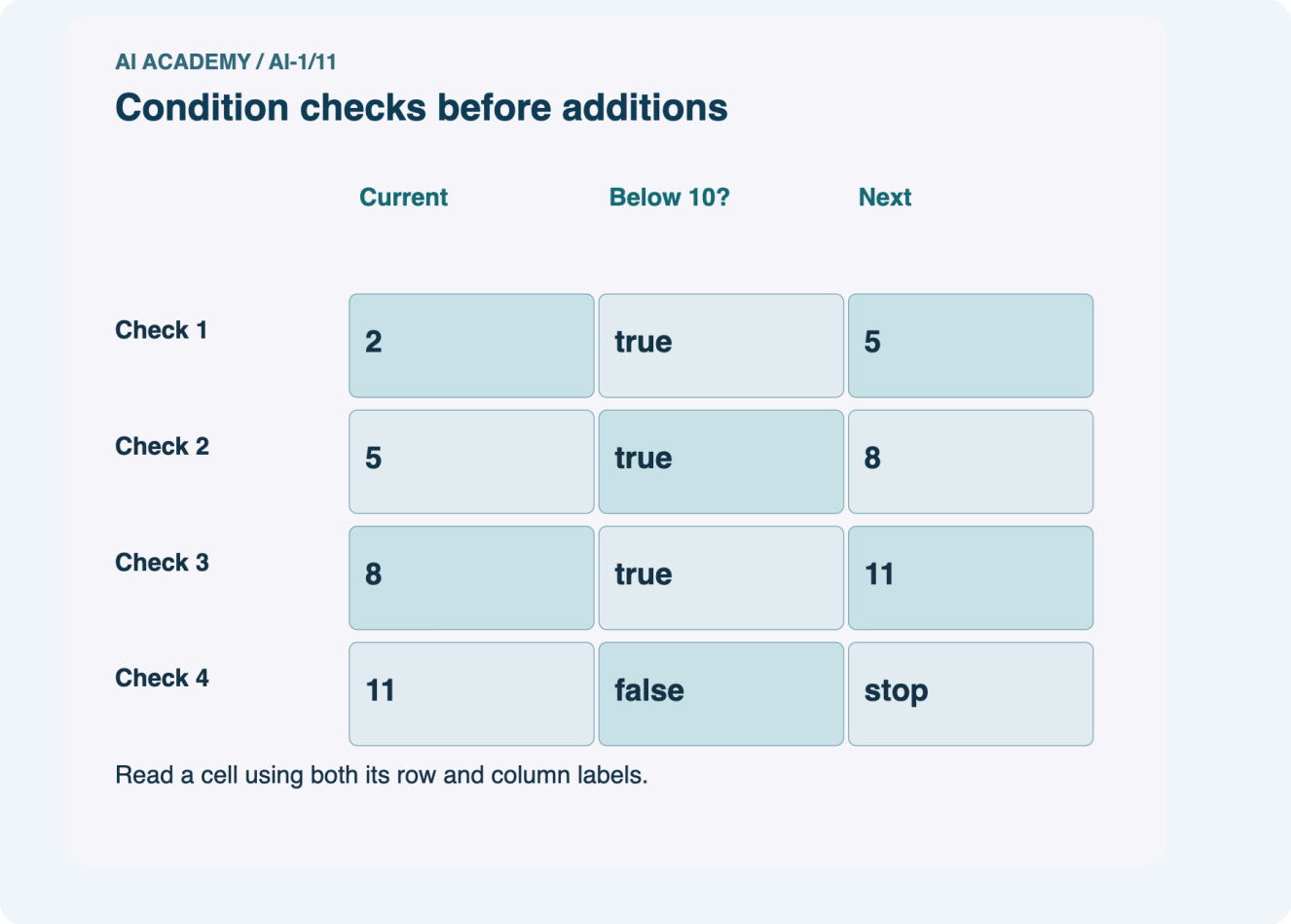
Follow the labelled picture with the matching explanation. Open the vector diagram

First test: 2<10 is true: update to 5. → Repeat: 5<10: update to 8. 8<10: update to 11. → Stop test: 11<10 is false; do not add again. → Report: Final counter 11 after three additions.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Condition checks before additions



Follow the labelled picture with the matching explanation. Open the vector diagram

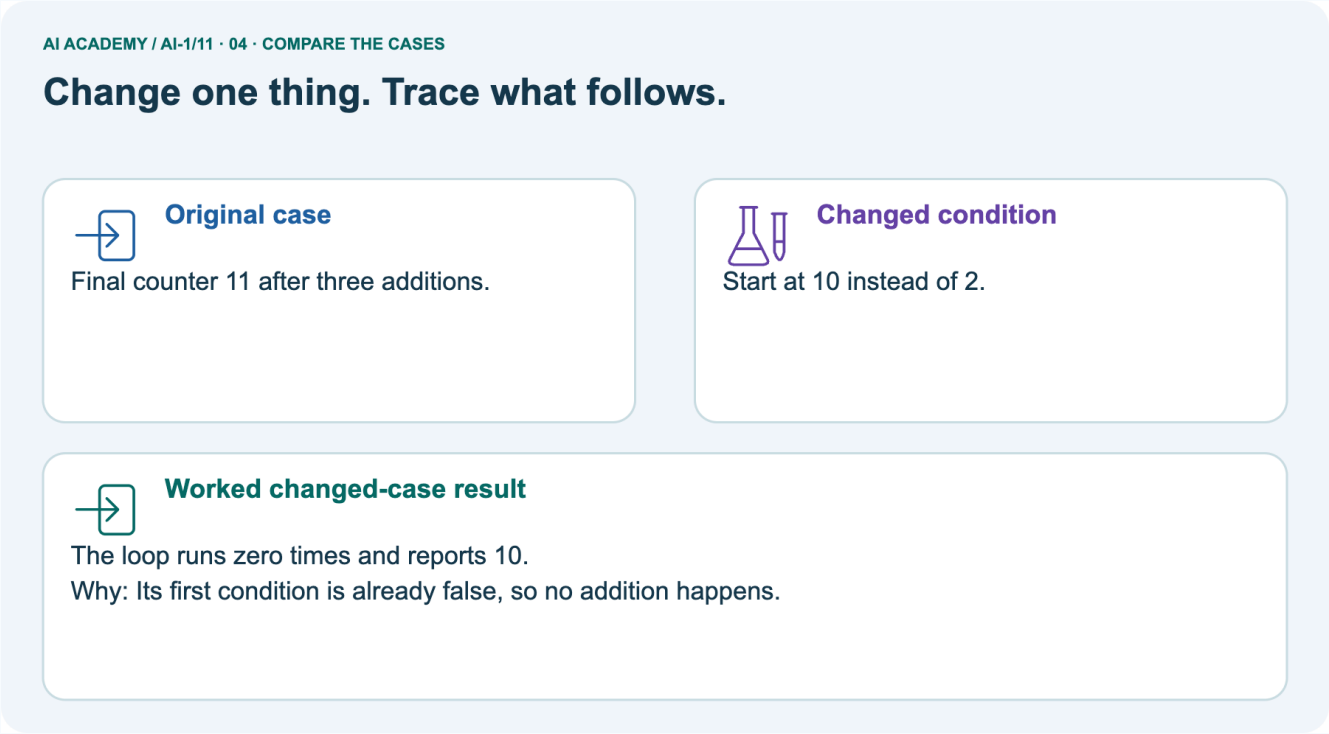
Condition checks before additions. Check 1: Current = 2, Below 10? = true, Next = 5. Check 2: Current = 5, Below 10? = true, Next = 8. Check 3: Current = 8, Below 10? = true, Next = 11.

Check 4: Current = 11, Below 10? = false, Next = stop.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

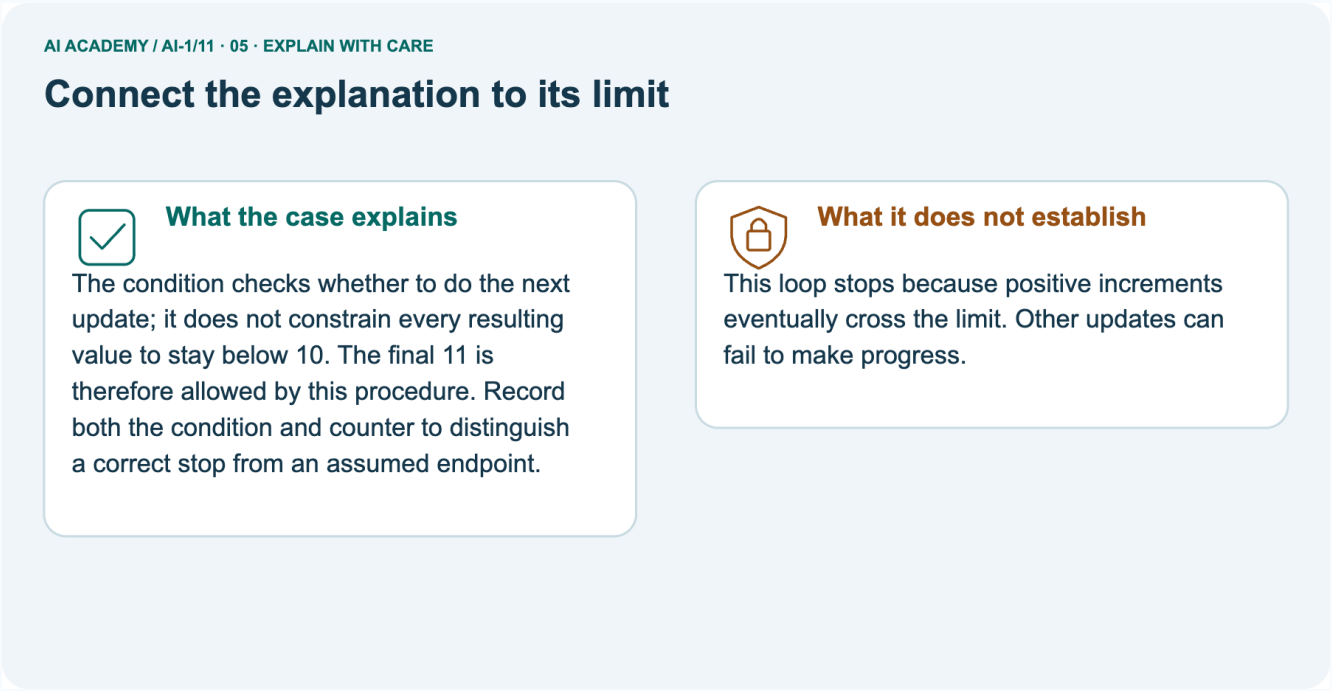
Original case: Final counter 11 after three additions. Changed condition: Start at 10 instead of 2. Expected result: The **loop** runs zero times and reports 10.

Reason: Its first condition is already false, so no addition happens.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

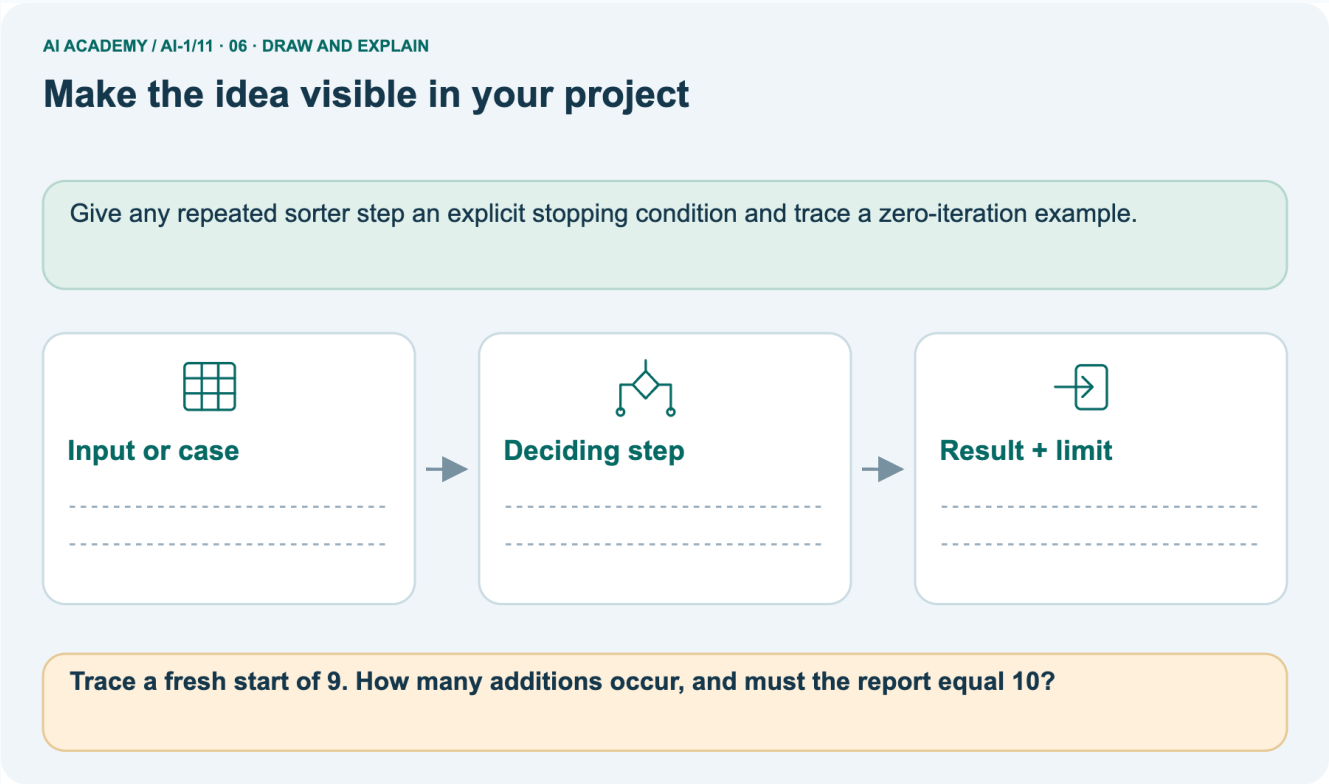
The condition checks whether to do the next update; it does not constrain every resulting value to stay below 10. The final 11 is therefore allowed by this procedure.

Record both the condition and counter to distinguish a correct stop from an assumed endpoint. Limit: This **loop** stops because positive increments eventually cross the limit. Other updates can fail to make **progress**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give any repeated sorter step an explicit **stopping condition** and trace a zero-**iteration** example. Draw your **input**, deciding step, result and limit. Supported question: Trace a fresh start of 9.

How many additions occur, and must the report equal 10?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Repeating Efficiently: the case

02. Follow the deciding steps

03. Condition checks before additions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.